

REEFPRINT/KHANYA: ore-hardness prediction on the belt and sulphide characterisation in the laboratory.

An analogue-data proof of concept and staged validation plan for South African concentrators

Mintek–SCi Grad Hackathon 2026 · Problem 3: Computer vision for real-time mineralogical characterisation

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Abstract

South African concentrators set mill feed, grind and reagents before the laboratory can say what was in the ore. REEF-PRINT/KHANYA combines a working microscope-image workflow with a separate hyperspectral hardness research track, and passes both to an operator who reviews every proposed change. All results so far come from public analogue datasets and a plant simulator. On 146 public porphyry-ore composites the hardness model reached an out-of-fold R^2 of 0.479 (colour alone 0.210), its nominal 90% upper bound covered 93.2% of samples, and a simulated feed-rate replay estimated a 1.7–1.9% throughput gain over a fixed design-feed policy. The historically evaluated microscope checkpoint reached IoU 0.870, 0.576 and 0.547 for pyrrhotite, chalcopyrite and pentlandite on 12 sections reused during development; the checkpoint in the audited application build differs, and an independent audit run on one section found large phase-abundance errors. Independent QEMSCAN agreement on unseen South African specimens, and a $\leq 10\%$ phase-abundance error, are validation targets, not results. We ask for a capped R210,000 laboratory and feasibility tranche; belt capital (a team allowance of US\$250,000–400,000, about R4.1–6.6 million, for one belt over six months) is released only after a separate belt-transfer gate. **[TECH NUMBERS PENDING: frozen model, independent results]**

1 Introduction

As Team Sonar, working on Mintek’s Problem 3, our responsibility is to give a concentrator’s shift team a reliable reading of its ore before that ore reaches the mill, and to turn the reading into a plant setting that a person can approve. This report sets out what we built, how we tested it, what it would cost and earn, and how staged local validation would test it on South African ore. It describes a research prototype: the results are on public analogue data and a simulated plant. It is written for metallurgists, mineralogists, process engineers and the people who fund their technology. Technical terms are defined in the glossary (Appendix A).

1.1 Problem statement

The brief states the problem plainly: “without real-time feedback, processing plants cannot adjust to changes in ore quality, leading to inefficient chemical usage and lower mineral yields” [1]. Ore reaches the mill within minutes of leaving the belt, whereas assays and automated mineralogy are batch laboratory work. Mintek’s own Minerals Processing and Characterisation Cluster runs QEMSCAN automated mineralogy [2], and a commercial laboratory describes mineralogy as “not an overnight assay service” [3]. Consequently, each shift sets the mill and the flotation circuit for ore whose mineralogy it has not yet seen.

The cost of deciding blind falls most heavily on South Africa. Figure 1 illustrates world platinum-group metal (PGM) mine production in 2025, highlighting that South Africa mined an estimated 120 of the world’s 170 tonnes of platinum (about 71%) and 70 of 190 tonnes of palladium. It also holds 63,000 of the world’s more than 76,000 tonnes of PGM reserves, mostly in the Bushveld Complex [4]. The same

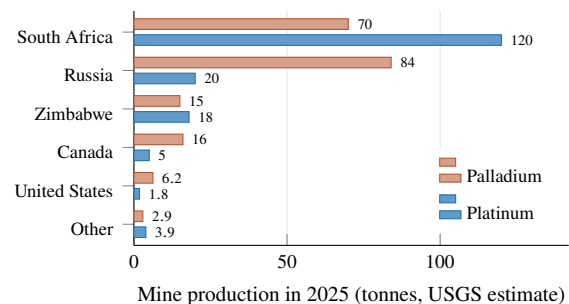


Figure 1: World platinum and palladium mine production in 2025. South Africa mined 120 of 170 tonnes of platinum; Zimbabwe’s Great Dyke ores are of the same family. Data: USGS [4].

source attributes the 9% fall in South African PGM output in 2025 to lower palladium prices, higher deep-level mining costs and disruptions to the electricity supply.

Recovery is where the money is. UG2 ore carries about 30% Cr_2O_3 against about 0.1% for Merensky ore, and the smelter caps the chromite in concentrate. Mintek testwork made UG2 concentrate at 87% recovery with 2.9% Cr_2O_3 , and recoveries above 90% were possible only if the Cr_2O_3 limit was relaxed [5]. Producers report recovery moves of this size to investors: Valterra raised concentrator recoveries by 1.2–1.7 percentage points at Mototolo and Amandelbult in 2025 [6]. Comminution, which sets both liberation and mill throughput, used 52% of site electricity in a benchmark of mines [7]. Plants do not decide entirely blind: operators, advanced process control (APC) and the mine plan already carry information about the ore. Thus, the decision we aim to improve is narrower: setting mill feed for each parcel of ore before it arrives, with a stated uncertainty, at a cost a producer can recover.

Table 1: The brief and the judging criteria: evidence and status. Shown = on public analogue data; Partial = part shown; Proposed = future work; Target = defined, not tested.

Requirement	Evidence (section)	Status
≥ 3 mineral phases	Pyrrhotite, chalcopyrite, pentlandite on analogue micrographs (4.4)	Shown
Accuracy report	Belt R^2 , bound coverage, baselines; microscope results per checkpoint (4.1–4.5)	Partial
$\leq 10\%$ error	Phase-abundance target defined; not yet tested on independent specimens (2.6)	Target
Adjust a plant parameter	Feed-rate proposal (designed); regrind advice to a simulator over OPC UA (2.3)	Partial
Real time	Hardness: 92 ms per parcel. Mineralogy: laboratory cadence, not on the belt	Partial
Local validation	South African ore, independent test sets (8)	Proposed
Innovation	Bounded hardness input that abstains, with operator review (2)	Shown
Feasibility	Staged plan, costs, support, compliance, risks (6–8)	Proposed
Impact	Scenario value per 1% more metal (6.4)	Scenario
Technical execution	Out-of-fold tests, baselines, failed ideas reported; 516 software tests (4)	Shown

1.2 Our solution: three speeds, one decision

REEFPRINT/KHANYA reads the ore at three speeds (Section 3):

1. **the belt** (REEFPRINT), every 30 seconds: a short-wave infrared (SWIR) camera above the mill-feed conveyor feeds a model that predicts each parcel’s grinding hardness with an upper bound, and Bond’s law turns the bound into a mill feed-rate proposal. The belt layer estimates hardness, not mineral identity;
2. **the laboratory** (KHANYA), once a shift: a microscope model classifies sulphide minerals in polished sections, which addresses the brief’s three-phase requirement on slower, prepared samples;
3. **the decision**: an operator reviews each proposed change before the ore reaches the mill; the site’s existing control system keeps authority, and every step is signed and recorded.

It is important to note what is not claimed: real-time mineralogy on the belt, performance on South African ore, a measured plant benefit, or identification of chromite and the silicates. Table 1 maps the brief and the judging criteria published in the hackathon FAQ [8] to the evidence, with the status of each.

2 How the system works

2.1 Data

No South African belt scans or labelled Bushveld micrographs were available, so every model was trained and tested on public data (Table 2). HIDSAG supplies hyperspectral scans of Chilean porphyry copper–molybdenum drill core with laboratory Bond work-index values [9]. LumenStone S2, a Norilsk nickel–copper sulphide ore [10], shares the mineral assemblage of Bushveld ores but not their abundance: sulphides cover 62.8% of its sections by area, whereas Merensky ore contains up to about 3% base-metal sulphides and UG2 less [5]. Nevertheless, it is the right teacher for the phases, because the base-metal sulphides in UG2 are predominantly pentlandite, chalcopyrite, pyrrhotite and pyrite [5].

Table 2: Datasets. All are public; none was supplied by Mintek or a producer.

Dataset	Ore	Size and use	Licence
HIDSAG [9]	Porphyry Cu–Mo, Chile	146 drill-core composites with Bond work index; visible (39 bands) and SWIR (75 bands). Hardness model	CC0
LumenStone S2 [10]	Ni–Cu sulphide, Norilsk	49 micrographs: 37 train, 12 held-out test. Microscope phases	Research use
Bushveld assays [11]	Bushveld chromitites	1,112 intervals. Check on PGE routing	CC BY 4.0

2.2 The belt layer: hardness from the infrared

Sensor (design). The design uses a SWIR line scanner such as the Specim SX25 (960–2500 nm, 640 pixels across, up to 162 lines a second) with a visible–near-infrared (VNIR) companion camera [12]. On a 1.2 m belt moving at 2 m/s it records a line every 12.3 mm with 1.88 mm pixels. It therefore measures parcels of ore, not individual particles. In the pilot design a *parcel* is 30 s of belt (about 60 m, or 4,860 scan lines averaged into one spectrum); in the replay reported here, a parcel is one HIDSAG composite scan of 80×117 pixels.

Model. Each parcel’s spectrum is reduced to features from the infrared absorption regions of clay, mica, chlorite and carbonate minerals (Fig. 4), and a model predicts the Bond work index (W_i , kWh/t). Results are reported only for samples the model never saw (out of fold), and a result counted only if it beat simple baselines under pass marks committed to the repository before each run. Five targets and several model variants were examined and only the work index survived, so the result is exploratory selection, not a confirmatory test; HIDSAG has no drill-hole identifiers, so grouped independence of the folds cannot be verified.

Bound and gate. The plant setting uses an upper bound that the true hardness should stay below 9 times out of 10 (a one-sided 90% split-conformal bound [13]), not the best guess. A more data-efficient bound [14] was tried first and failed its pass mark, so the exact bound that the plan named as its fallback is used, and the result is provisional. The 90% is a marginal coverage target that assumes new parcels resemble the calibration data; it is not a per-parcel guarantee and does not certify protection against overload. An out-of-distribution (OOD) check flags parcels unlike anything in training, and their advice is withdrawn.

2.3 The decision layer and its cadence

The decision layer turns the hardness bound into a mill setting that a person can approve. Ball mills are usually limited by motor power, so the tonnes a mill can grind depend on how hard the ore is. Bond’s law gives the specific grinding energy [15]

$$W = 10 W_i \left(\frac{1}{\sqrt{P_{80}}} - \frac{1}{\sqrt{F_{80}}} \right) \text{ kWh/t}, \quad (1)$$

where F_{80} and P_{80} are the feed and product sizes (μm) passed by 80% of the material. At a fixed grind target the feed rate the mill can carry is proportional to $1/W_i$, so the proposal is

$$F_{\text{proposed}} = F_{\text{design}} \times \frac{W_{i,\text{design}}}{W_{i,\text{bound}}}. \quad (2)$$

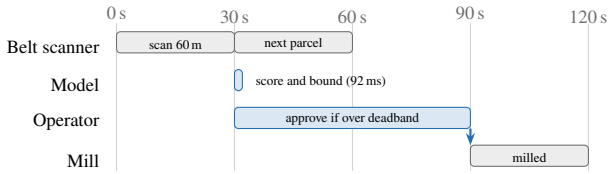


Figure 2: Proposed decision cadence for the pilot, to be agreed with the site. A parcel is scored as soon as it is scanned; the operator then has the rest of the assumed 90 s belt-to-mill transit to approve a change. Once a shift, the metallurgist approves the envelope, ramp limits and deadband. No approval in time, or a flagged parcel, means the advice is withdrawn and the site’s control system continues.

The ratio removes explicit power and size terms only when power, feed size, product target and circuit efficiency are held constant; the pilot must test that approximation for the selected circuit. For example, a design hardness of 16.6 kWh/t and a bound of 16.9 kWh/t give 98.2% of design feed. The proposal must sit inside an operating envelope set by the site’s metallurgist; the demonstration envelope is 85–110% of design feed. The demonstration falls back to 100% of design feed; at a site, the fallback is the host’s own operating policy under its existing controls.

Figure 2 illustrates the proposed cadence, highlighting that a person approves changes, not every computation. A new proposal is computed every 30 s, but the operator sees it only when it differs from the running setpoint by more than a deadband (for example 2% of design feed) set by the metallurgist; smaller changes are not made. The operator may approve, modify within the envelope, reject or escalate. If nobody approves before the parcel reaches the mill (an assumed 90 s from the end of the scan), or the parcel is flagged, or the data are stale, the advice is withdrawn and the site’s own control policy continues; the research system never raises feed on its own. Feed may be cut by up to 10% but raised by only 3% per parcel. At most 120 proposals an hour are computed, but only changes larger than the deadband reach the operator, so far fewer are expected. The replay data have no time order, so the shadow phase measures the actual operator-facing rate before anyone acts on it. Every event is written to a hash-chained record with conventional (Ed25519) and post-quantum (ML-DSA-65 [16]) signatures; this supports traceability, not the correctness of the measurement. The microscope’s advice reaches a simulated regrind setting over OPC UA [17] (simulated); the feed-rate path is designed, not built.

2.4 The laboratory layer: minerals under the microscope

The microscope model is a standard image-segmentation network (DeepLabV3 with a ResNet-50 backbone [18, 19]) that labels every pixel of a reflected-light image as resin, pyrrhotite, chalcopyrite, pentlandite or magnetite. Only the approved checkpoint (SHA-256 de7135a9, full hash in Appendix C) may send advice to the simulator. A *Quick* scan classifies six 512-pixel fields of a section; a *full* scan covers the whole section.

Connected regions of the mask are split into grains. For this demonstration, chalcopyrite and pentlandite are treated as payload and pyrrhotite as reject; that assignment is a demonstration assumption for this ore, because pyrrhotite’s value depends on PGM deportment and the host’s treatment objec-

tives. A grain whose area is at least half payload is *payload-dominant* (shown as “free” in the application), otherwise *reject-dominant* (“locked”). This is a 2D payload-area proxy, not a measure of liberation or recoverability; the 50% line is a declared demonstration choice. A rule-based advisor turns these measures into *grind finer*, *continue*, *verify before acting* or, when the rejected phase dominates, *adjust reagent*. Each is a hypothesis for a metallurgist to test, not an instruction. It answers *verify* when the payload-area estimate lies within ± 0.335 of the 50% line, and gives no recommendation with fewer than 9 payload-bearing grains or mean confidence below 0.85. The ± 0.335 band was calibrated on the 12 evaluation sections (Section 4.5); the other thresholds are placeholders for site metallurgists to set.

2.5 How we tested it

Belt models were scored out of fold against the strongest simple baseline (the fold mean, the fold median and a colour-only model), with pass marks committed to the repository before each run. The microscope was scored on the publisher’s 12 S2 test sections. They were excluded from network training but reused during development, to compare pipelines, select the checkpoint and calibrate the advisor band, so we report them as *development-influenced evaluation sections*, not an independent test. An independent, source-disjoint local test set is the first deliverable of the laboratory study. Accuracy is the intersection over union (IoU: the overlap between predicted and expert-labelled areas, from 0 to 1), pooled over 103.8 million pixels and averaged per section, with bootstrap confidence intervals over sections. Pixels are correlated within a section, so sections, not pixels, are the independent units. Decision-level agreement runs the same advisor on the model’s mask and on the expert’s mask; a disagreement is *unsafe* if the model gives a confident instruction that the expert mask does not support. This measures sensitivity to segmentation error, not whether the rules themselves are metallurgically right.

2.6 Validation targets and independent protocol (proposed)

The $\leq 10\%$ error target must be defined before samples are collected, because absolute and relative errors differ: a reference of 2 wt% predicted as 3 wt% is a 1-point absolute error but a 50% relative error. We propose that the mineralogist and a statistician sign an acceptance contract naming the decision-critical phases, an absolute tolerance below an agreed quantification limit and a $\leq 10\%$ relative target above it, and whether the target applies per specimen or as a phase-wise mean. Microscope image fractions are 2D areas, whereas QEMSCAN modal results are usually mass fractions, so both bases must be reported and converted explicitly.

A blind test needs more than hiding answers: a separate custodian holds specimen identities and laboratory results; the frozen model receives only acquisition inputs and permitted metadata; predictions are hashed before the reference is revealed; splits are grouped by specimen and collection event; and every refusal stays in the denominator. As a starting envelope we propose 30 development, 10 calibration and at least 30 locked evaluation specimens, subject to a power calculation and laboratory quotes. Zero failures in 30 independent specimens bounds the failure rate at about 9.5% (one-sided 95%), which is a statement about pass/fail, not a guarantee of 10%

numerical error. If a model fails, it is fixed on development material and must face a new untouched cohort. [TECH NUMBERS PENDING: signed protocol, sample counts, metric definitions]

3 System architecture

Figure 3 illustrates the system, highlighting that both layers pass the same gates and the same operator review. The belt and laboratory layers run on different time scales: laboratory results give slower context and audit evidence, not a check on each 30 s parcel. The belt monitor (REEFPRIENT Live) runs the hardness model, the decision view and the signed record. The microscope workbench (KHANYA) is a web application (React) on a Python service (FastAPI and PyTorch) that runs on an ordinary laptop processor. In local mode it accepts connections only from the same machine and makes no external requests, so a network outage at the plant does not stop the advice. No language model computes a measured value or a control value.

Speeds were measured on a four-core laptop processor: 92 ms per parcel (median); 16.7–21.2 s per six-field microscope scan on mains power (5 runs) and 30.8 s on battery. A static public demonstration runs the belt monitor on pre-computed data; a secure server passed 17 of 17 security tests locally and is not deployed. Software verification is in Appendix C.

4 Results

4.1 Belt: predicting grinding hardness

Of the five HIDSAG targets tested (copper and molybdenum recovery, lime consumption, pH and Bond work index), only the work index beat its strongest baseline under every pass mark. Out of fold, the hyperspectral model explains 48% of the variance in work index (R^2 0.479, mean absolute error 1.05 kWh/t), a model given only colour explains 21% (R^2 0.210, 1.29 kWh/t), and the fold mean and median explain nothing (R^2 -0.027 and -0.041), so the infrared carries most of the information. Mean measured W_i is 14.62 kWh/t.

Figure 5 illustrates how the bound behaves, highlighting that it was exceeded on 10 of 146 samples (6.8%). The 95% interval for coverage (88.4–96.6%) spans the 90% target, so the result is compatible with it but does not establish future coverage, particularly on new ore. The price of the bound is width: the bound sits on average 1.64 kWh/t, about 11% of the mean hardness, above the best guess. The OOD check passed 137 parcels, marked 5 borderline and refused 4. Figure 4 shows one parcel as the model sees it. For parcel GMET-0004 the predicted W_i is 15.9 kWh/t and the bound 16.9 kWh/t, against a design hardness of 16.6 kWh/t (the training data's 90th percentile), so the proposal is 98.2% of design feed. The laboratory value was 12.1 kWh/t, so the bound was conservative here.

4.2 Belt: from hardness to a plant decision

Figure 6 illustrates four feed-rate policies replayed over all 146 composites against a fixed design-feed policy (the 90th-percentile hardness), highlighting the trade-off between throughput and overloads. In the replay a parcel is *overloaded*

when its true hardness exceeds the hardness the feed was set for; this is a model quantity, not a plant trip. Setting the feed from the bound, with OOD parcels withdrawn, gave a simulated throughput gain of +1.9% (95% CI +0.9 to +2.9%), and the overload share was 6.8% against 8.9%; the one-sided 95% upper bound on the difference (+1.4 points) is inside the +3-point margin in the evaluation plan. The original replay did not apply the 85–110% envelope; a later analysis with it gives +1.7% (+0.7 to +2.7%) and the same overload share. Feeding from the best guess instead would add 13.6% but overload 45.2% of parcels, which is why the policy uses a bound. Perfect knowledge would add 14.0%, so the bound captures about one-seventh of what is available: its width (1.64 kWh/t) is where the rest is lost, and a tighter bound from site data is the main route to a larger gain.

The result is provisional for three reasons: the exact bound was adopted after the more efficient one failed, HIDSAG has no time order (so transit time, stockpile mixing and the asymmetric ramp limit are not simulated), and the ore is not Bushveld ore. Symmetric ramp limits were tested over 200 random parcel orders: $\pm 10\%$ per parcel left overloads unchanged (-0.3 points), but $\pm 5\%$ added 2.1 points and $\pm 2\%$ added 4.7, so feed must be cut fast. A robustness test defined before it was run found that a new capture of the same ore keeps its error ($\times 1.00$), as do half the field of view, added noise and $\pm 15\%$ illumination; a wavelength drift raised the error by a factor of 1.25 unflagged, so a wavelength-calibration check was added to the design.

4.3 Is it physically possible?

Every claim was tested against its physics, and the failures stay in the report. Bond's law gives the throughput gain above under its stated assumptions (provisional). In the evaluated flotation scenarios, grinding faster cost up to 0.25 percentage points of recovery on the steep part of the recovery curve and none on the plateau; this is the largest loss among the scenarios modelled, not a physical bound, and the site's recovery curve must be measured. Re-ordering ore during load-shedding cannot create tonnes, because energy per tonne is conserved over a cycle: it gave -3.8% in simulation, and even perfect knowledge gave -3.0%, so that claim was rejected. The camera's 12.3 mm lines and 1.88 mm pixels resolve parcels, not particles; no test showed that the belt camera can identify minerals, so the microscope carries the phases; and the system is optical only, with no radioactive source and no radiation sensing.

4.4 Laboratory: phase identification

The microscope addresses the brief's three-phase requirement on analogue ore. Figure 7 illustrates its accuracy by class, highlighting IoU values of 0.87 for pyrrhotite, 0.58 for chalcopyrite and 0.55 for pentlandite (precision and recall in Appendix B). The pooled mean IoU is 0.5725 (95% CI 0.494–0.624); averaged per section it is 0.4671 (0.412–0.534). The model clearly beats trivial baselines (majority class 0.117; colour-only 0.395).

Magnetite, a fourth class the brief does not require, is never predicted. It is only 0.8% of the test pixels: 78% of it is labelled as resin, 17% as pyrrhotite and 5% as pentlandite. The last share is about 0.3% of all pixels predicted as pentlandite, so magnetite barely moves the payload measures; where it is

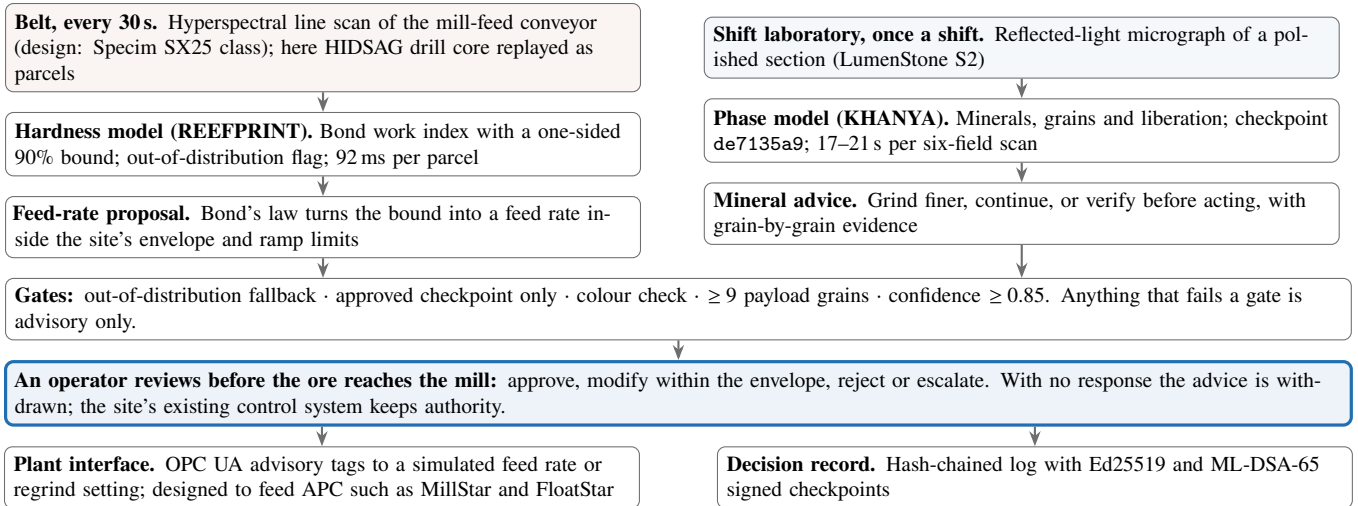


Figure 3: System architecture: three speeds, one decision. The belt layer (left) is the site system; the microscope layer (right) is the laboratory truth layer and delivers the brief's three phases. Both pass the same gates and the same operator review. Built: the models, applications and signed record (on replayed public data). Simulated: the plant and regrind path. Designed: the belt sensor and feed-rate path.

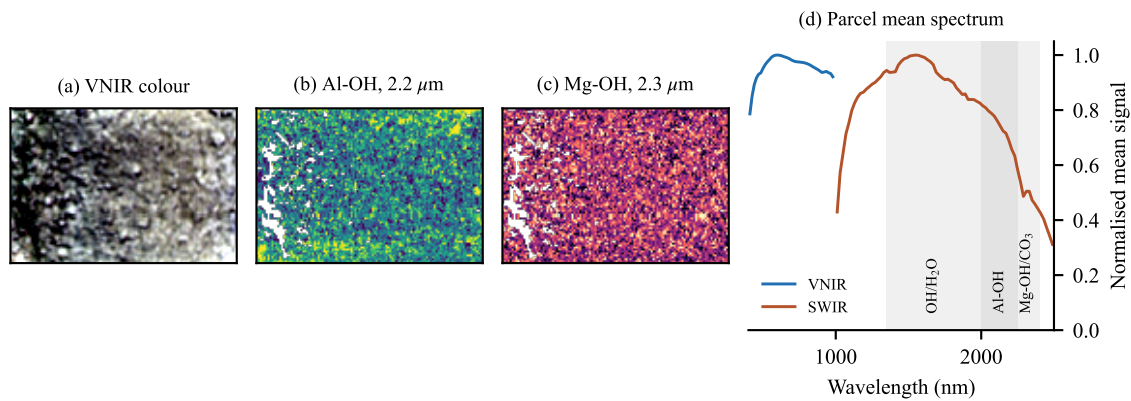


Figure 4: One belt parcel as the hardness model sees it: HIDSAG composite GMET-0004 (Chilean porphyry Cu–Mo), 80×117 pixels. (a) Colour image from the visible bands. (b, c) Relative absorption depth at the Al–OH (white mica, clays) and Mg–OH (chlorite, biotite, carbonate) features; detector stripes removed for display only. (d) Mean spectrum, with the absorption regions the model uses. The model predicts W_i 15.9 kWh/t with a 90% upper bound of 16.9 kWh/t (laboratory value 12.1 kWh/t). The maps show absorption features, not a mineral identification.

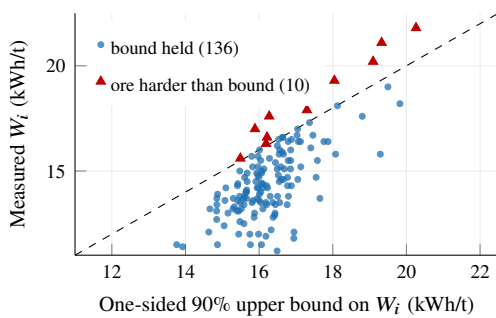


Figure 5: Calibration of the hardness bound, out of fold, 146 composites (no drill-hole grouping available). Points below the dashed line are samples the bound covered: 93.2% (95% CI 88.4–96.6%) against the 90% target. The bound sits on average 1.64 kWh/t (range 1.04–2.14) above the prediction.

intergrown with a sulphide grain, dropping it slightly overstates that grain's payload share.

Three checkpoints exist and must not share scores (Table 3). The figures above describe de7135a9. The application build audited on 6 October ran fb78727d, which the application itself marks as not approved. On one previously used section

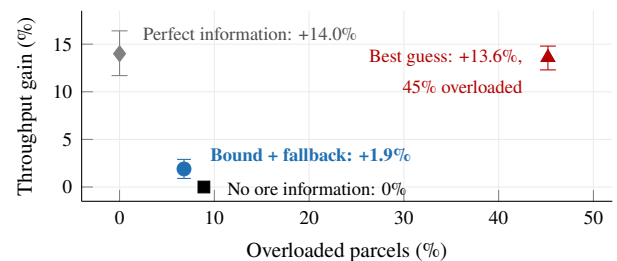


Figure 6: Feed-rate policies replayed over 146 composites against a fixed design-feed policy (simulated; not PGM ore, not a time series, original replay without the envelope), with 95% confidence intervals. Up is more throughput; left is fewer replay overloads.

(test_11, six fields), the audit compared that model's output with the expert masks: pixel accuracy was 64.1%, pyrrhotite was predicted at 61.1% of the area against 90.7% in the reference, and pentlandite at 34.8% against 2.0%. This is one development section, not QEMSCAN and not an independent cohort, but it is a concrete counterexample to any blanket $\leq 10\%$ claim for the active model. A retrained candidate, 42646cfa, scores 0.632 overall and detects magnetite, but loses pyrrhotite and pentlandite and has pentlandite precision of only 0.50 (Ap-

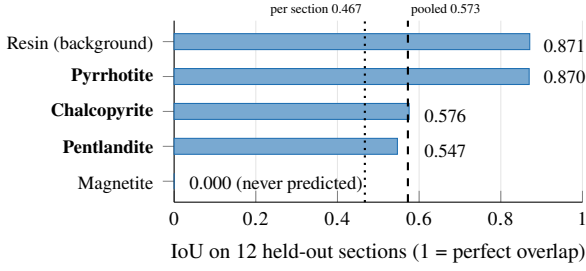


Figure 7: Microscope accuracy by class for the historically evaluated checkpoint de7135a9, full-section inference on development-influenced sections. The three phases the brief requires are in bold. Section means range from 0.340 to 0.728 (Appendix B).

Table 3: Model registry. Scores are bound to one checkpoint each. **[TECH NUMBERS PENDING: frozen version and independent results]**

Checkpoint	Evidence	Status
de7135a9	Mean IoU 0.573 pooled, 12 development-influenced sections	Historical evaluated; research demo
fb78727d	Mean IoU 0.454 (re-run); audit on test_11: pixel accuracy 64.1%, pentlandite 34.8% vs 2.0%	Active in audited build; not approved
42646cfa	Mean IoU 0.632; pentlandite IoU 0.446, precision 0.50	Candidate; quarantined

Table 4: Decision-level confusion on the 12 held-out sections: advice from the expert mask (rows) against advice from the model’s mask (columns). The advisor can also flag a reagent issue when the reject phase dominates; this happened once, on the expert mask only.

Expert mask \ model	Grind	Continue	Verify	Too few	Total
Grind finer	3	0	3	2	8
Continue	0	1	2	0	3
Adjust reagent	0	0	1	0	1
Total	3	1	6	2	12

pendix B); it is quarantined, and the comparative decision safety of the checkpoints is untested.

4.5 Laboratory: decisions and grain evidence

Table 4 compares the advice from the model’s masks with the advice from the expert masks. The model produced 4 definite recommendations, 6 *verify* outputs and 2 *too few grains* outputs on the 12 sections, so its action coverage was 4 of 12 (33%). No definite recommendation disagreed in the unsafe direction with the same advisor applied to expert masks. This is a small, development-influenced rule-agreement result: it does not validate the rules themselves, establish an independent unsafe-action rate or demonstrate plant safety. The one-sided 95% upper bound of 22.1% on 0 of 12 refers to all sections, not to risk when the system acts. Both the rules and the abstention thresholds need independent evaluation on the host stream. The counts are for de7135a9. **[TECH NUMBERS PENDING: frozen-model decision results]**

It is important to note that the ± 0.335 verify band was calibrated on these same sections, so the agreement result is partly in-sample. Replaying the liberation rule on the stored section measurements shows how much it depends on the band: on the 9 sections it governs, the result stays at 0 wrong for bands of ± 0.25 or wider (7 correct at ± 0.25), but one section (test_04)

becomes unsafe below that, and two with no band at all. The band must therefore be re-set on independent sections in the laboratory pilot. A realistic exposure shift (-35 levels on each colour channel) changed the advice on 8 of 12 sections, so illumination control is also a pilot requirement.

The application’s grain inspector lets a mineralogist audit the advice grain by grain; its “free/locked” labels overstate a four-phase 2D classifier, and relabelling them as high or low apparent target-phase fraction is planned, not implemented. In Quick mode each field is classified without context beyond its edge; on one example, test_11, 19 of 24 grains come within 6 pixels of a field edge, so Quick-mode grain statistics are indicative until a border policy is set and checked.

5 Comparison with literature and limitations

5.1 Comparison with literature

Published work supports the plausibility of the approach, not its validity here. Notole et al. [20] validated hyperspectral imaging and portable XRF against reference methods on Merensky Reef drill core scanned at the Council for Geoscience in Pretoria; that shows the sensing works on Bushveld material, not that our hardness model does. The three sulphides the microscope classifies are the main base-metal sulphides of UG2 ore [5], and our 50% proxy line coincides with a boundary between middling classes in Mintek’s liberation scheme [2], although that boundary does not by itself define free material. Mintek’s AI optimisation raised a lead–zinc concentrator’s throughput by 0.5%, worth over US\$2 million (about R33 million) a year [21], which shows that moves of this order are valued, not that this system would deliver them.

Four ideas were tested and withdrawn, and they mark where the method stops. The belt spectra did not predict QEM-SCAN mineralogy once size fraction and process line were accounted for; a later grouped re-analysis of the same HID-SAG MINERAL1 data (99 records, 36 composites, already used in development) confirmed this: all records were within 10 absolute points, yet only 23–91% of eligible records were within 10% relative error depending on the phase, and a simple process-line baseline beat the spectral model for chalcopyrite and pyrite. The belt claim is therefore limited to hardness. Belt chemistry routed Bushveld PGE ore worse than the mine plan’s seam label (balanced accuracy 0.79 against 0.82). The belt did not predict recovery, lime or pH better than a baseline. Plant tags never forecast concentrate silica an hour ahead better than the last laboratory assay.

5.2 Limitations

The following limitations apply, and (a), (b) and (e) block any investment decision until tested: (a) no South African ore has been evaluated, and a model trained on Chilean porphyry core may not transfer, because hardness also depends on silicification, grain size and fracturing; (b) the belt result is a replay of 146 composites without time order or drill-hole grouping, against a fixed design-feed policy rather than the plant’s existing control, and a camera sees the belt surface, which may not represent the mass entering the mill; (c) the belt camera predicts hardness and does not identify minerals, and the microscope has not been trained on chromite or the silicates; (d)

the 12 microscope sections guided development, and the approved checkpoint cannot be regenerated exactly; (e) the advisor's rules (including the pyrrhotite-reject and 50% assumptions) are unvalidated for any host ore, its band was set on the evaluation sections, and liberation is a 2D proxy; and (f) latency is measured on one laptop, the plant is a simulator, and economic benefit has not been attributed at any site; and (g) there is no fresh paired PGM validation, the active application checkpoint differs from the evaluated one, chromite and silicate classes are missing, and reference-laboratory uncertainty is not yet quantified. Compared with the abstract Mintek accepted, three things changed: the five Bushveld phases became three sulphides on an analogue ore, because no labelled Bushveld images existed; the QEMSCAN "teacher" became the pilot's truth-laboratory stream; and the fallback was reversed from holding the last setpoint to applying the conservative setting.

6 Business model and financial case

6.1 Value proposition and business model

Our value proposition is simple: the plant gets a reading of its ore before the mill, with a stated bound, and after the trial it pays only for value that an independent party has measured. No producer, Mintek or TIA has agreed to fund, host or license the work; the roles below are proposals. The buyer is the concentrator manager or general manager for metallurgy at a South African PGM producer, who owns recovery, concentrate quality and mill throughput; producers such as Valterra, Implats, Sibanye-Stillwater, Northam and Tharisa are market segments, not customers. Figure 8 illustrates how money and value flow, highlighting the two kinds of payment: the pilot is paid at cost, and only the later fee depends on measured value. The Technology Innovation Agency (TIA), the national agency that funds technology development towards commercialisation [22], is a possible funder of the local validation stage, subject to its programme criteria.

The proposed commercial structure has three parts: a fixed, capped validation contract for the research work; a separately agreed service fee after acceptance; and an optional variable fee capped at 20% of independently verified net value, with the baseline, measurement period, adjustments, negative-value treatment and audit rights defined in advance. Willingness to pay must still be tested through buyer interviews; laboratory savings are not claimed, because 4 definite decisions in 12 sections do not show that any paid analyses can be dropped. The model fits Mintek's published work [21]: the models need QEMSCAN composites and Bond tests, which is laboratory work; a hardness input could add ore-side information to process control that today works from process data; Mintek's benefit-sharing model with Hindustan Zinc has generated over R35 million since 2021; and its products reach over 60 countries through distributors such as Molycop and Mercantil. South African PGM output also fell 9% in 2025 [4], while Valterra's realised basket price rose from R32,611 per ounce in 2025 to R45,993 in the first half of 2026 [6, 23].

Table 5 compares the options a plant already has. The system replaces none of them; whether it adds information beyond existing APC, operator practice and mine-plan blending is what the local trial must show.

Table 5: What plants use today, and where this system differs.

Option	What it gives	Where this differs
Cross-belt neutron analysers	Bulk elements, minute by minute	Hardness, not elements; isotope sources need a SAHPRA licence [24], this needs none
Slurry analysers	Grade and Cr_2O_3 in slurry, after milling	Reads the ore before the mill; their output is an input
Automated mineralogy (QEMSCAN)	Phases and liberation, in days	Predicts between truth results; its composites retrain the models
APC (e.g. MillStar, FloatStar)	Stable, optimised circuits from process data	A partner: an ore-side input inside APC limits

6.2 Pricing, and what the money pays for

Table 6 sets out one scoped budget, which replaces the earlier R8.2–49.2 million installation grid; that grid survives only as a superseded sensitivity in Table 7. The first request is a capped R210,000 laboratory and feasibility tranche. Its allocation is a proposed control, not a supplier estimate, and it does not promise that 70 specimens can be characterised within it: reference work is paid only against an itemised quote. Belt capital is a separate team allowance released only after the belt-transfer gate (G1B). VAT, duties, major civil work, conveyors or sorters, shutdown costs and closed-loop control are excluded and would be funded separately. After a trial, annual support is assumed at US\$30,000–60,000 (R0.49–0.98 million), not added again to the six-month support line.

6.3 From throughput to recovered metal

The belt result is a throughput gain, and revenue comes from recovered metal, so the two must be bridged. For a consistent stream, metal output is proportional to dry tonnes $\times$ head grade $\times$ recovery, so

$$\frac{\Delta M}{M} = \left(1 + \frac{\Delta T}{T}\right) \left(1 + \frac{\Delta G}{G}\right) \left(1 + \frac{\Delta R}{R}\right) - 1, \quad (3)$$

where T is throughput, G head grade and R recovery. With unchanged grade, a 1.75% throughput gain and a recovery loss of 0.25 percentage points from 87% (a relative change of -0.29%) give 1.46% more metal. Where the plant is limited by ore supply, extra throughput is worth nothing. If all of Zondereinde's output passed the instrumented circuit (an upper bound), 1.46% more metal would be worth R64–123 million a year in margin, and a quarter of that gain R16–31 million; these are simulated gains, not measurements.

6.4 Financial methodology

The following assumptions apply: (a) annual output is that of Northam's Zondereinde mine, 333,050 equivalent refined 4E ounces in F2026 [26]; (b) prices are Valterra's realised basket prices of R32,611 per ounce (2025) and R45,993 (first half of 2026), with cash operating costs of R19,488 and R20,677 per ounce [6, 23], used as proxies for another company's mine, with different baskets, payability and cost structure; (c) for a commercial-life case, installed cost is the belt allowance without the six-month support line (US\$175,000–280,000, R2.87–4.59 million) and running cost is the post-trial support (R0.49–0.98 million a year); and (d) the equipment has a 5-year life at

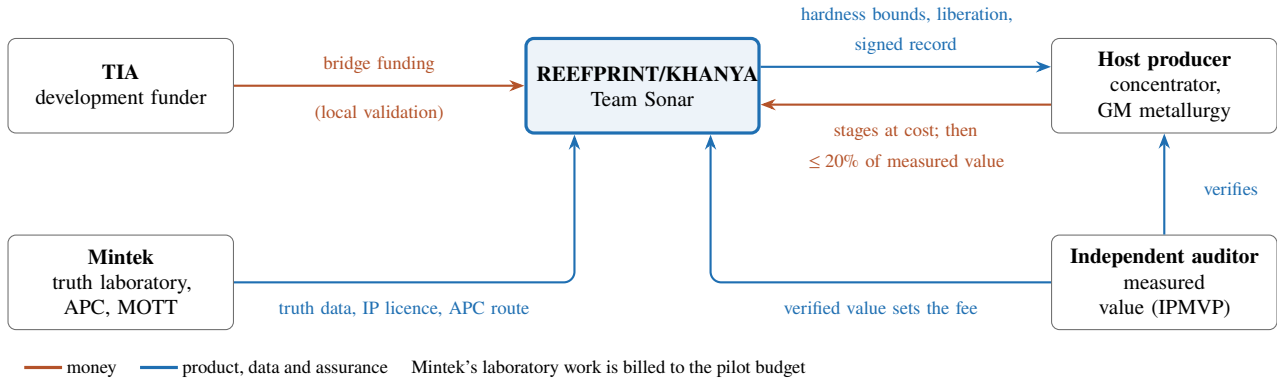


Figure 8: The proposed business model; partner roles are proposals, not agreements. The producer would pay for development stages at cost; after a successful trial it would pay a capped share of independently measured value, so no measured value means no fee. Mintek would earn from the laboratory work the models need; licensing follows the hackathon IP agreement with its Office of Technology Transfer (MOTT).

Table 6: One scoped budget: proposed allocations and team planning allowances, not quotations. Rand at the illustrative R16.4 per US\$ (January–September 2026 average [25]).

First tranche: laboratory and feasibility (proposed caps)			One belt, six months (team allowance, after G1B)	
Item	Rand cap	Evidence before payment	Item	US\$ (rand)
Protocol and method design	15,000	Agreed specimen unit, endpoints	Sensors, optics, references	100–160k (R1.64–2.62 M)
Sampling, logistics, custody	25,000	Host schedule, sample register	Lighting, enclosure, mounting	25–40k (R0.41–0.66 M)
Preparation and repeatability	20,000	Laboratory quote: mounts, splits	Edge computer, UPS, network	10–15k (R0.16–0.25 M)
Reference laboratory work	65,000	Itemised quote, minimum dataset	Installation, integration, commissioning	40–65k (R0.66–1.07 M)
Data, model and evaluation work	35,000	Versioned inputs, outputs, code	Sampling, reference, support (6 months)	25–40k (R0.41–0.66 M)
Independent review, close-out	15,000	Signed evidence review	25% contingency	50–80k (R0.82–1.31 M)
Controlled reserve (20% of base)	35,000	Documented risk event	Total	250–400k (R4.10–6.56 M)
Total	210,000	No overrun without a new decision	With first tranche	R4.31–6.77 M

a 10% discount rate; and (e) an affected-output fraction f of the mine's output passes the instrumented circuit, taken as 1 unless stated. The annual value of an uplift u in recovered metal is

$$V = u \times f \times Q \times m, \quad (4)$$

where Q is annual output in ounces and m the margin per ounce. For $u = 1\%$ and $m = \text{price less cash cost}$,

$$V = 0.01 \times 333,050 \times (\text{R}13,123 \text{ to } \text{R}25,316), \quad (5)$$

which is equivalent to R43.7–84.3 million a year (R108.6–153.2 million in revenue before costs). Deducting the full average cash cost may be conservative where the mill has spare capacity, but extra mining, handling or bottleneck costs could change that, so a host-approved incremental cash-flow model is needed before investment. The installed cost is spread over the equipment's life with the capital recovery factor

$$\text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1} = 0.2638 \quad (r = 10\%, n = 5), \quad (6)$$

so the annualised cost is

$$C = \text{CRF} \times C_{\text{installed}} + C_{\text{running}}, \quad (7)$$

which is equivalent to R1.25–2.20 million a year, or R2.50 million with the 25% contingency on the high case. The break-even uplift is then $u_{\text{BE}} = C/(f \times Q \times m)$. Table 7 shows

Table 7: Break-even sensitivities in relative recovered metal (recalculated under the stated assumptions; illustrative, not a site funding threshold). One percentage point of recovery at 80% recovery is 1.25% relative.

Scenario	More metal needed
Low cost, 2026 margin, all output affected	0.015%
High cost, 2025 margin, all output	0.050%
High cost + 25% contingency, 2025 margin	0.057%
High cost, 2025 margin, 20% fee before system costs	0.063%
High cost, 2025 margin, half of output affected	0.10%
High cost, 2025 margin, a quarter of output affected	0.20%
High cost, 30% lower basket price (2025 costs)	0.20%
Superseded R49.2 M installation grid, 2025 margin	0.52%

how it moves with cost, margin, affected fraction and the fee basis. If the 20% fee is charged on the margin before system costs, the producer breaks even only when $0.8V = C$. Figure 9 illustrates the all-output case.

Annualising over five years describes continued successful use, not whether a nine-month experiment pays for itself. The pilot cash view is simpler: if the laboratory or belt-transfer gate fails, the loss is at most the R210,000 first tranche, because no belt equipment is ordered before G1B; if the belt stage proceeds and then stops, the exposure is up to R4.31–6.77 million, with equipment resale assumed to be zero in the

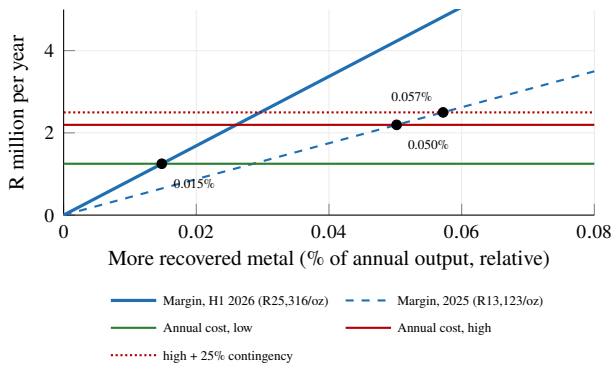


Figure 9: Break-even if all of Zondereinde’s output were affected ($f = 1$): annual margin on extra recovered metal (blue) against the annualised commercial-life cost of the one-belt allowance (horizontal lines), before any fee. A pilot that fails does not reach this case.

pessimistic case. The fee basis must be agreed so that no support cost is charged twice, and measured value needs an agreed metallurgical accounting boundary and attribution method under IPMVP [27].

6.5 Risks, contingency and fallbacks

Table 8 sets out the main risks of using the product, as the team assesses them. Each has a mitigation that acts before it happens and a fallback that limits the damage if it does. Contingency pays only for uncertain costs within the approved scope, not for adding sites, sorters or closed-loop control. Three fallbacks run through all of the risks: the plant’s existing controls keep authority; each stage ends at a gate where spending can stop; and the fee is zero if no value is measured. Fallbacks reduce exposure to detected failures; they do not guarantee safety against undetected ones.

7 Implementation and support

The pilot depends on people more than on hardware. Appendix B gives the detailed plans; this section summarises them.

Governance (Table 12). A host metallurgist would own the decision envelope; a reference laboratory the characterisation method; an independent custodian the locked test identities and results; and the team the versioned software and reproducible analysis. The host owns plant operation and stop authority, and the initial site configuration is read-only. These are required functions, not appointments: one person may cover several, but independent assessment and spending approval must not collapse into self-certification. Within the team, two roles are filled by the authors and one (software, security and deployment) is to be recruited or sourced in kind.

Maintenance (Table 14). Calibration and data-quality checks withdraw advice when a specified failure is detected. They reduce exposure to known failure modes but cannot detect every harmful change; undetected drift, sampling mismatch and wrong rules remain trial risks.

Compliance (Table 13). These are planned controls, not achieved compliance. The Mine Health and Safety Act requires a risk assessment with the health and safety committee and training before significant changes to plant or equipment [28]; POPIA governs approvers’ names in the record [29]; Mining Charter clause 2.3.1 counts skills and R&D spend

[30]; IP from public funding falls under the IPR Act [31]; Mintek and TIA spend under the PFMA [32]; and the camera needs no radiation licence. Before installation, the health and safety committee, the management-of-change review, the IT/OT change board and the capital committee sign off.

Independent fact-checkers to outsource: (a) a SANAS-accredited laboratory for Bond tests and assays [33]; (b) Mintek’s QEMSCAN for mineralogy; (c) an independent auditor for the value on which fees are charged, using IPMVP [27]; (d) an IEC 62443 assessor for operational-technology security [34]; (e) an IP attorney through MOTT; and (f) a university reviewer for the model audit.

Training (Table 15). Each group receives a short module before its phase starts, aligned with the site’s workplace skills plan, which mines report to the Mining Qualifications Authority [35].

8 Roadmap: evidence before capital

Figure 10 illustrates the plan, highlighting that the transfer question is answered before any belt equipment is bought. The microscope (G1A) and belt-transfer (G1B) gates are separate: a mine can accept the laboratory tool while rejecting belt prediction, and microscope success alone never releases belt spending. A useful first 90 days could end with “we identified a narrower valid use and stopped an unjustified purchase”; we propose that the funder recognise that as a successful de-risking outcome. Table 9 lists what each gate must show and who decides; the numerical thresholds are to be agreed with the host before data are collected, so that the study can return a clear negative result.

9 Conclusions and recommendation

The present contribution is a functioning, inspectable evidence workflow and a falsifiable route to local validation. On public analogue data the hardness model improves on simple baselines and the microscope classifies three relevant sulphide phases, but the feed-rate benefit is a simulation, the microscope results are development-influenced, the active application checkpoint has shown large abundance errors on one section, and the $\leq 10\%$ error target is untested. We therefore recommend a capped R210,000 laboratory and feasibility tranche with quote-backed gates, and belt capital only after a separate belt-transfer gate. Investment should buy resolution of the remaining measurement, transfer and operational uncertainties; a commercial claim will depend on independent reference agreement, useful acceptance coverage and measured net value on the affected stream. **[TECH NUMBERS PENDING: final model results]**

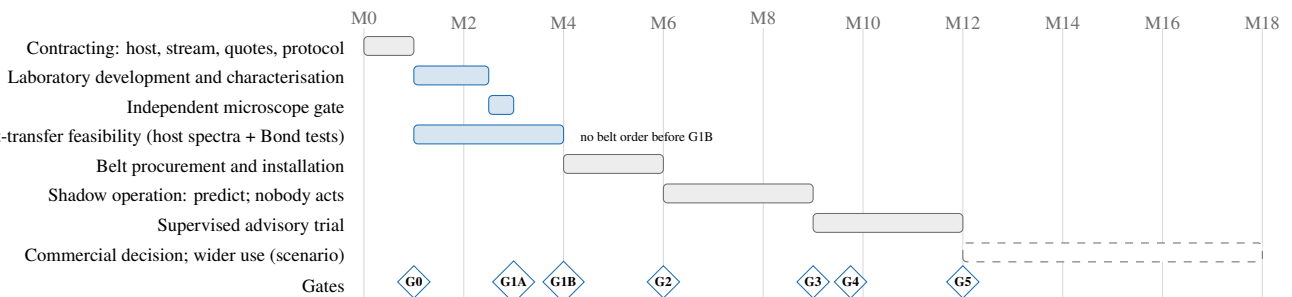
CRedit authorship contribution statement. **Lethabo Hoaeane:** Conceptualisation, domain lead, belt hyperspectral modelling, belt monitor and decision application, economics, presentation. **Sibusiso Khumalo:** Microscope segmentation model, accuracy and decision-level evaluation, advisor and safety gates, OPC UA simulation, workbench review and verification, technical report.

Declaration of competing interest. The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 8: Risk register: mitigation before the event, and the contingency or fallback if it happens.

#	Risk	Mitigation (before)	Contingency or fallback (if it happens)
R1	The hardness model does not transfer to Bushveld ore	Belt-transfer feasibility on host samples (G1B) before any belt order	Stop after the first tranche; no belt spend
R2	Belt conditions: dust, moisture, vibration, lamp ageing, wavelength drift	Purge air, calibration references and a calibration gate at installation acceptance (G2)	Detected faults withdraw advice; undetected drift remains a risk, checked by monthly audits
R3	Power outages and load-shedding	Runs offline on an edge computer with a UPS	After a restart, advice is blocked until checks pass
R4	Cyber attack or an unsafe write to the plant	Read-only advisory tags; no write path to safety systems; IEC 62443 zones; signed records	Disconnect; the plant continues on its existing control system
R5	Operators ignore, over-trust or cannot keep up with the advice	Deadband and shift-level approvals; training before the advisory phase; named roles	Advice expires; site control continues
R6	No measurable gain	On/off operating blocks with confidence intervals; independent measurement and verification	Stop at G5; no fee is due
R7	Sensor costs 25% more, or the rand weakens 10%	Comparable quotes before G1B; 25% belt reserve (US\$50–80k)	Sensor overrun of US\$25–40k uses half the reserve; a 10% weaker rand adds R0.41–0.66 M, so reforecast before ordering
R8	A fall in PGM prices	The fee is a share of measured net value	The fee falls with value automatically
R9	Funding delay, host withdrawal, or staff not available (two of three team roles filled)	The first tranche is small; named deputies; third role recruited or sourced in kind	Pause capital commitments; preserve lawful evidence; qualify a replacement host
R10	Wrong advisor rules, or rights to use the training data commercially	Rules validated by a mineralogist on host samples; LumenStone is licensed for research use, so commercial rights are clarified with the rights holder	Display measurements without prescriptive advice; retrain on host-owned data

**Figure 10:** Proposed staged roadmap, in months from an agreed readiness event rather than a funding announcement (planning estimate; sample supply, test turnaround, sensor lead times and site windows can extend it). Shaded bars are the evidence-first stages; dashed bars are scenarios. Diamonds are go/no-go gates (Table 9). Readiness levels are awarded from demonstrated evidence, not elapsed time.**Table 9:** Proposed gates. Numerical thresholds are agreed with the host before evaluation data are collected.

Gate	Release condition	If it fails
G0	Named host, stream and owners; quotes; data and IP terms; signed protocol	Fund preparation only
G1A	Independent microscope agreement on locked host specimens, with useful acceptance coverage	Research image review only
G1B	Host spectra beat a metadata baseline for hardness, with agreed coverage and bound width	Stop the belt track
G2	Sensor quote, factory and site acceptance, risk assessment, read-only design	Repair or remove
G3	Reliable timing, representative paired sampling, drift and refusal behaviour	Extend or stop
G4	Host approves envelope, change management, training, rollback, trial plan	Revert to host procedure
G5	Independently measured net value, total cost, host acceptance	Close out; no fee

Use of AI tools. Code and drafting were assisted by AI coding assistants. Every reported number was computed by code in the project repository and checked by the authors.

Data and code availability. HIDSAG is available under CC0 (doi:10.6084/m9.figshare.c.5983921) [9]; LumenStone under research-use terms [10]; the Bushveld assays under CC BY 4.0 (doi:10.17632/dc8jcnbcvk.1) [11]. Code, checkpoint hashes and

every results file behind the numbers in this report: <https://github.com/Sibusiso-K/KHANYA> (Appendix C). A static demonstration of the belt monitor runs at <https://lethabomh14-reefprint.static.hf.space>.

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A Glossary

APC	Advanced process control, e.g. Mintek’s MillStar and FloatStar
Bond work index (W_i)	Energy needed to grind ore, in kWh/t; harder ore has a higher W_i
Bound	Here, a hardness the true value should stay below 9 times in 10 (one-sided 90% conformance bound)
Fallback	In the demonstration, 100% of design feed; at a site, the host’s own operating policy under its existing controls
Deadband	The smallest change worth showing the operator
Envelope	The range of feed rates the site’s metallurgist allows (demonstration: 85–110% of design)
Payload-dominant	A grain at least half payload by 2D area (“free” in the app); a proxy, not liberation
IoU	Intersection over union: overlap of predicted and expert areas, 0 to 1
Parcel	30 s of belt in the pilot design; one composite scan in the replay
PGM, 4E	Platinum-group metals; 4E = Pt, Pd, Rh and Au
QEMSCAN	Automated scanning-electron mineralogy
SWIR, VNIR	Short-wave and visible–near infrared light

B Supporting tables

The final page gives microscope per-class results and pixel confusion (Tables 10–11) and the accountability, compliance, maintenance and training plans (Tables 12–15). Per-section mean IoU for de7135a9 (test_01 to test_12): 0.413, 0.728, 0.659, 0.466, 0.472, 0.340, 0.453, 0.504, 0.375, 0.378, 0.377, 0.443.

C Software verification and reproducibility

Tests. 181 unit-test functions and 7 browser tests (workbench) and 328 (belt software); these do not certify physical performance. *Checkpoints* (SHA-256 prefixes; full values in the repository’s reports/). de7135a96541a46d (historical evaluated), 42646cfafbeac539 (candidate), fb78727d4859947d (active in the audited build). *Release.* Audited application release: commit 85061d4. A release manifest binding all artefacts is proposed. *Reproducibility.* Inference reproduces from archived weights; retraining does not: a second run of the de7135a9 recipe scored 0.454 because the patch sampler was unseeded. It is now seeded (adeb1ee).

Table 10: Microscope per-class results, approved checkpoint against the re-trained candidate, 12 S2 test sections (full-section inference).

Class	Approved de7135a9			Candidate
	IoU	Recall	Precision	IoU (precision)
Resin	0.871	0.920	0.942	0.848 (0.960)
Chalcocopyrite	0.576	0.716	0.746	0.800 (0.937)
Magnetite	0.000	0.000	–	0.248 (0.255)
Pyrrhotite	0.870	0.934	0.926	0.818 (0.950)
Pentlandite	0.547	0.742	0.675	0.446 (0.503)
Mean	0.573			0.632

Table 11: Pixel confusion for the approved checkpoint (% of each true class), 103.8 million test pixels.

True \ predicted	Resin	Ccp	Mag	Po	Pn
Resin	92.0	0.8	0.0	5.7	1.5
Chalcocopyrite	1.6	71.6	0.0	3.1	23.7
Magnetite	78.3	0.1	0.0	16.8	4.8
Pyrrhotite	1.1	1.6	0.0	93.4	3.9
Pentlandite	0.7	0.8	0.0	24.3	74.2

Table 12: Accountability for the pilot (required functions, not appointments).

Role	Accountability	Minimum evidence
Host sponsor or plant manager	Site participation, resources, production priorities, stop authority	Signed pilot charter and named deputy
Host metallurgist	Use case, operating envelope, fallback policy, interpretation of results	Approved protocol, baseline and trial limits
Independent mineralogist or laboratory	Reference method, phase taxonomy, quality control	Method statement, quote, uncertainty, signed reports
Data custodian and statistician	Blind split, endpoint lock, independent evaluation	Access separation, hashed manifests, analysis plan
Team technical lead (Team Sonar)	Software and model delivery, reproducibility, incident fixes	Release registry, tests, support cover, rollback
OT and control engineer, safety lead	Installation risk, network boundary, change approval	Risk assessment, read-only design, factory and site acceptance
Finance, procurement and IP owners	Spending, contracts, asset, IP and data rights	Quote comparison, cash forecast, agreements, registers

Table 13: Laws, policies and standards that apply, and the planned controls (not achieved compliance).

Law, policy or standard	What it asks of this project	Planned control (evidence required)
Mine Health and Safety Act 29 of 1996 [28]	Assess risks of new equipment, consulting the health and safety committee (s. 11, s. 34); train employees before significant changes to plant or equipment, and keep records (s. 10)	Installed under the site's risk assessment and lockout; advisory only; operators trained before the advisory phase
Protection of Personal Information Act 4 of 2013 [29]	Lawful, limited processing of personal information, such as approvers' names	Purpose, access, retention and non-disciplinary use agreed with the information officer before the shadow phase
Mining Charter 2018, clause 2.3.1 [30]	Invest 5% of the leviable amount in essential skills development, including R&D in processing and technology efficiency	Pilot training and R&D may count toward it; clause 2.3 was not among the clauses set aside by the High Court in 2021 [36]
IPR from Publicly Financed R&D Act 51 of 2008 [31]	IP from publicly financed R&D is identified, protected and commercialised for South Africa (NIPMO)	IP managed by MOTT under the hackathon agreement, with invention credits [8]
Public Finance Management Act 1 of 1999 [32]	Mintek (Schedule 3B) and TIA (Schedule 3A) spend public money under PFMA controls	Funding released in tranches tied to gates G0–G5, with auditable deliverables
Hazardous Substances Act 15 of 1973 (SAHPRA) [24]	Licences for radioactive sources (Group IV)	Not applicable: optical sensing only
IEC 62443 [34]; OPC UA [17]	Zones and conduits for industrial cyber security; secure data exchange	Read-only advisory tags; no write path to safety systems; IT/OT sign-off
ISO/IEC 17025, accredited by SANAS [33]	Demonstrated competence of testing laboratories	Bond tests and assays used as truth come from accredited laboratories
IPMVP (EVO) [27]	A recognised protocol for measuring and verifying savings	Fees invoiced only on value measured under an agreed plan by an independent party

Table 14: Proposed maintenance plan; the site sets the final intervals.

Task	Interval	Owner
Reference checks; refuse if a feature shifts over half a band	Every reference cycle (automatic)	System
Window cleaning and purge-air check	Each shift, or on alarm	Instrument technician
Lamp replacement	Lamp-life schedule	Instrument technician
Enclosure temperature and dew point	Continuous alarm	System; electrical
Microscope colour and illumination check	Every image; daily	Laboratory technician
Model audit against Bond tests and QEMSCAN	Monthly in the pilot	Metallurgist, mineralogist
Retraining	Only through a validation gate	Team Sonar
Security patches; backup restore test	Monthly; quarterly	Team Sonar, IT/OT

Table 15: Proposed training plan. Records are kept as the MHSA requires.

Who	What they learn	Format; when
Control-room operators, supervisors	Reading a proposal; approve, modify, reject or escalate; the deadband; what the fallback does	Half a day plus supervised shifts; before G4
Metallurgists	Setting the envelope and deadband; reading bounds and liberation; reviewing the record	One day; at G0
Laboratory staff	Workbench, colour calibration, grain audit, comparison with QEMSCAN	One day; before G1A
Instrument technicians	Camera, enclosure, references, lamps, lockout	One day with the vendor; at G2
IT and OT staff	Network zones, backups, patching, incident response	Half a day; before installation
Managers and funders	Reading the measured-value report and the gates	One hour; each gate